ENGINEERING PORTFOLIO · 2026

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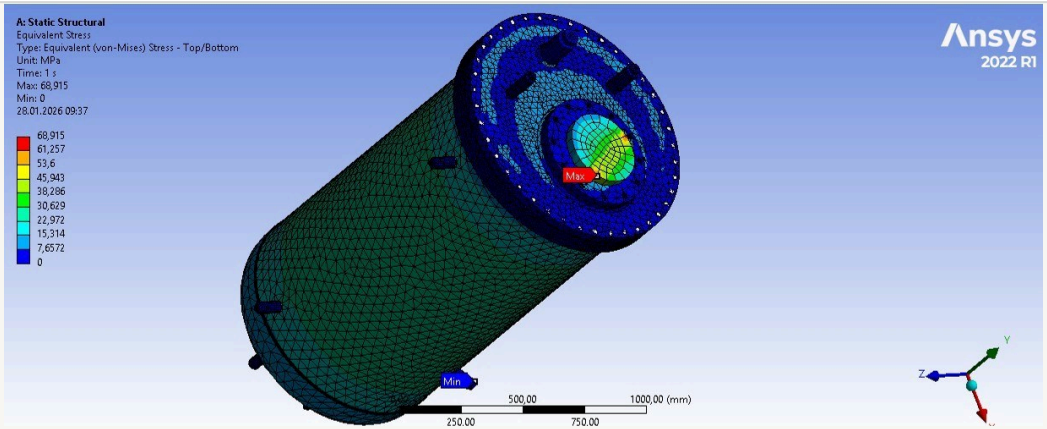
01 · STRUCTURAL ANALYSIS

Pressure vessel and bolted head: where is the design governed?

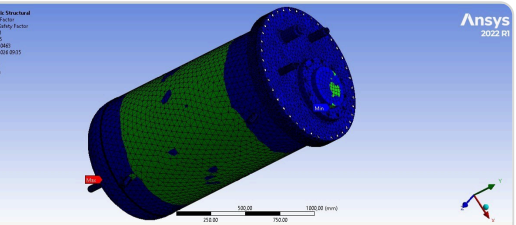
Mechanical System Design project, İÜC
Advisor: Dr. Murat Bozkurt · individual work

A cylindrical pressure vessel with a bolted head under internal pressure. Shell theory gives a comfortable safety factor for the shell. The question was whether that number holds once the real geometry of the head and flange is modelled.

11.2	3.05	68.9 MPa	0.597 MPa
safety factor, shell membrane (theory)	minimum safety factor, FEA	peak von Mises at the head	internal pressure



Equivalent (von Mises) stress. The peak sits at the head and flange, not in the shell.



Safety factor map, minimum 3.05

PROBLEM

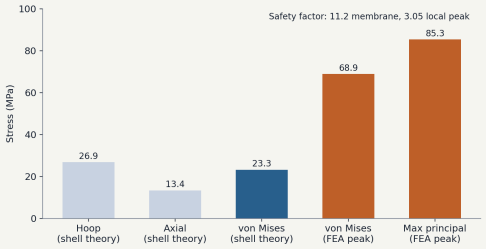
Shell r = 450 mm, t = 10 mm, head 12 mm, L = 1600 mm, SA-516 Gr.70 (yield 260 MPa), internal pressure 0.597 MPa. Find the stresses, the deformation and the real margin against yield.

APPROACH

- **Hand calculation:** thin-wall shell theory for hoop, axial and von Mises stress, radial growth (0.051 mm) and the head membrane stress.
- **FEA:** static structural model in ANSYS Mechanical 2022 R1, linear elastic, pressure on the inner faces, mesh refined at the flange and head.
- **Comparison logic:** theory describes the shell away from the ends, FEA peaks include bending and notch effects at discontinuities. I compared them at two levels instead of forcing one number onto both.

RESULT

The shell membrane is far from yield (factor 11.2), but the head and flange region governs the design with a minimum factor of **3.05**, consistent with yield over the peak principal stress (260 / 85.3).



Shell theory against FEA peak values (data from the project report)

WHAT I WOULD DO NEXT

A mesh independence study at the peak, a minimum-constraint boundary condition instead of the fixed base, and a radial displacement check at mid-shell against the 0.051 mm hand result.

01 · CONTINUED

From stress to safety distance

Article in preparation with Dr. Murat Bozkurt

A safe stress level at working pressure does not say how far people should stand if the vessel ruptures. I extended the same vessel with a consequence chain: stored energy, TNT equivalent and blast overpressure distance.

0.81 MJ	0.19 kg	14.5 m	45.8 m
stored energy at working pressure	TNT equivalent	0.5 psi distance, working pressure	0.5 psi distance, burst case

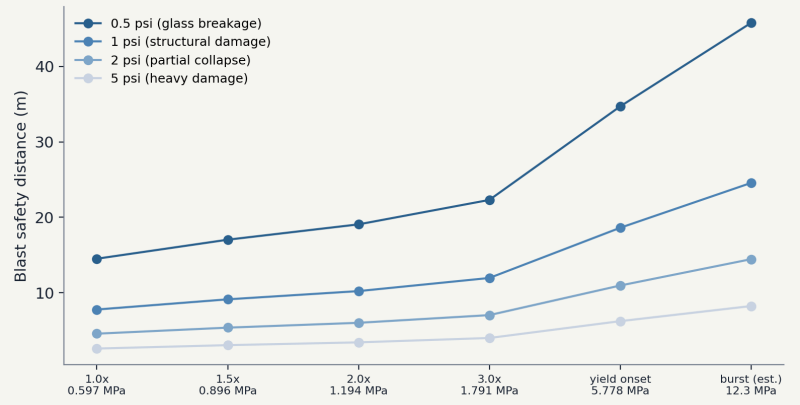
METHOD

- **Stored energy:** isentropic expansion of an ideal gas (k = 1.4), internal volume about 1.1 m³.
- **TNT equivalent:** energy divided by 4.184 MJ/kg, with the 4.68 MJ/kg bound reported as uncertainty.
- **Distance:** Hopkinson-Cranz scaling, $R = Z \cdot W^{1/3}$, with Z for 0.5, 1, 2 and 5 psi from the Kinney-Graham free-air correlation.
- **Scenarios:** 1x to 3x working pressure, yield onset (about 5.78 MPa from shell theory) and an estimated burst pressure of 12.3 MPa.

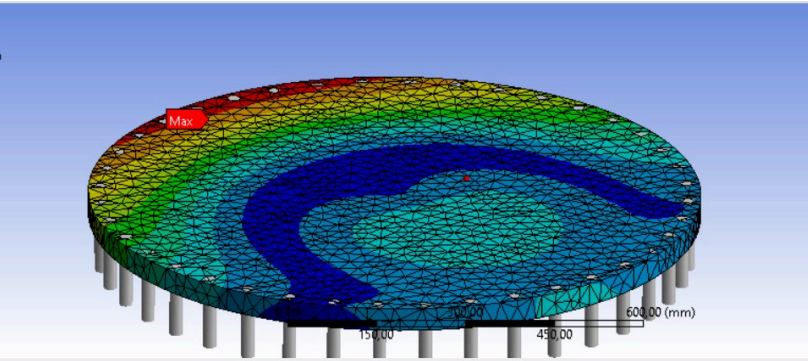
WHY IT MATTERS

The results were turned into layered safety zones on a conceptual site layout. The study also states its limits clearly: fragment throw, which often governs real burst events, was discussed qualitatively because the data needed to calculate it was not available.

> Academic pre-engineering assessment, not a site safety study. Fluid type, volume and burst pressure are stated assumptions.



Blast safety distance by pressure scenario and overpressure threshold (V = 1.1 m³)



Bolted head model used in the study, total deformation

02 · DESIGN AND MANUFACTURING

TEKNOFEST UAV: a skid landing gear that survives the landing

TEKNOFEST International UAV Competition · Aug 2024 - Aug 2025
Team project · my part: landing gear

A 7 kg X-frame quadcopter needs a landing gear that is light, stiff enough to keep the camera clear and tough enough to absorb a hard landing. I designed the skid gear and its printed joints.

7.05 kg	2.32 kg	292 mm	50%
take-off mass of the UAV	frame mass	landing gear height	below the specification weight target



Full UAV assembly in SolidWorks



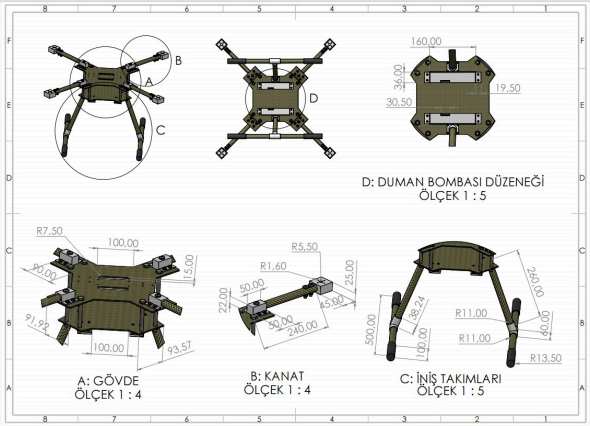
Skid legs: carbon tubes, printed T-joints and rubber pads

DESIGN DECISIONS

- **Skid instead of arm-mounted legs:** keeps the gear out of the camera's view and spreads the landing load over a larger contact area.
- **Carbon-fibre tubes** (5 mm wall on the legs, thicker than the arms) for stiffness at low mass, sourced from a local manufacturer.
- **Printed T-joints and angled flanges:** ABS was chosen for its strength-to-weight and easy printing. Rubber pads on the skids reduce the landing shock.
- **Easy to fit and remove:** the gear is bolted to the frame, like the other parts, for easy assembly.

TOOLS

SolidWorks (parametric parts, assembly and drawings), ANSYS static structural analysis of the landing gear, FDM printing in PLA, ABS and ASA.



Drawing sheet: frame, arm and landing gear dimensions

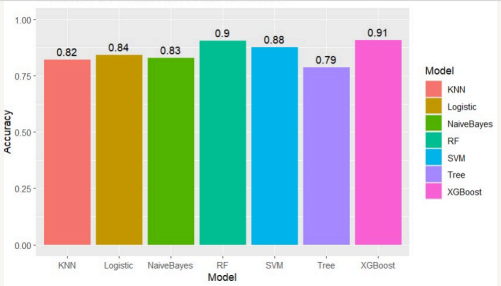
03 · DATA

Predicting steel plate defects before they reach the customer

Machine learning course project · Feb 2025 - Jun 2025
Individual work · R

Seven fault types appear on steel plates in production. The goal was a classifier that tells them apart from the inspection measurements, so that defective output can be caught and traced back to the process.

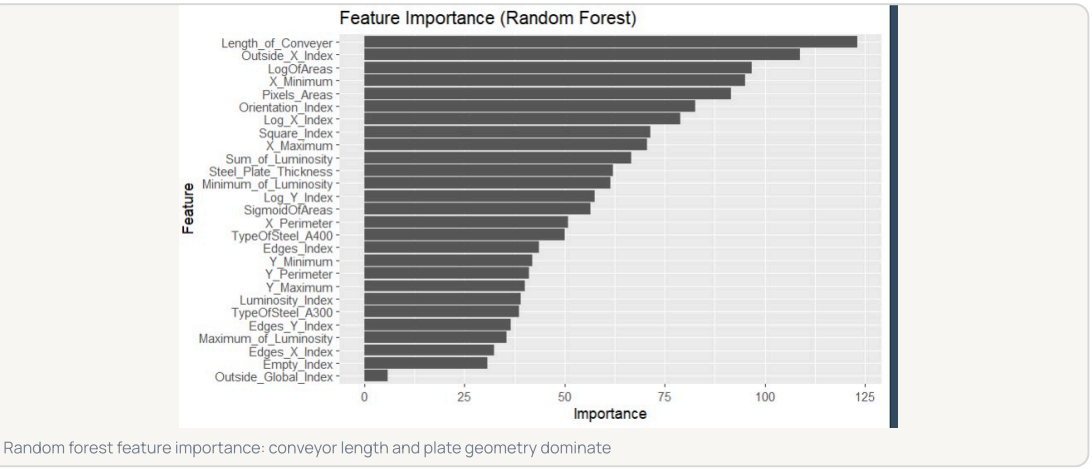
91.3%	0.895	7	27
voting ensemble accuracy	macro F1-score	models compared	input features



Test accuracy of the seven models



Confusion matrix (XGBoost)



Random forest feature importance: conveyor length and plate geometry dominate

DATA AND PREPARATION

UCI Steel Plates Faults data, 27 geometric and luminosity features per plate. The ambiguous "Other faults" class was left out, the data was split 80/20, scaled, the minority classes were oversampled and Boruta selected the informative features.

MODELS

Logistic regression, decision tree, random forest, k-NN, naive Bayes, SVM (radial kernel, 5-fold cross-validation for C) and XGBoost. XGBoost (91%), random forest (90%) and SVM (88%) were combined by majority vote, which reached **91.3%** with macro precision 0.909 and recall 0.891.

WHAT THE MODEL SAYS ABOUT THE PROCESS

K-scratch and bumps are recognised well, while Z-scratch is often confused with other classes. Conveyor length ranks highest, followed by plate position and area features, so the model relies mostly on line and geometry measurements.

04 · MORE WORK

Formula Student and industry

İÜC RACING FORMULA STUDENT · POWERTRAIN TEAM · 2023 - 2026

Designed the chain guard in SolidWorks and verified its strength with static analysis in ANSYS. Revised the bracket design for a **15% weight reduction**.

INTERNSHIPS

- **FESTO**, 2025: traced mechanical, pneumatic, electrical and software faults in a robotic system and recommissioned it.
- **BEKO**, manufacturing, 2025: found loss sources at 5 critical assembly stations and proposed 2 improvements.
- **Midas Makina**, design, 2024: revised SolidWorks models and prepared manufacturing drawings to tolerancing standards.
- **Yılmaz Redüktör**, workshop, 2024: CNC turning, milling, heat treatment, grinding and gear quality control on site.

SKILLS

SolidWorks (CSWP), ANSYS Mechanical, AutoCAD, Siemens NX, CATIA, Fusion 360, MATLAB/Simulink, R, MS Excel. Static and modal FEA, topology optimisation, technical drawing and GD&T, design for manufacturability, design for injection moulding, sheet metal design, root cause analysis, lean manufacturing.

CONTACT

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